Formal Lab Report – Unknown “A”

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Introduction

This report will aim to identify an unknown compound via analysis of nuclear magnetic resonance spectroscopy, infrared spectroscopy, and gas chromatograph-mass spectrometry. The ability to identify unknown compounds via such tests provides an incalculable benefit in the realm of chemical analysis and identification, in both laboratory and real-world settings.

Body

Nuclear Magnetic Resonance Spectroscopy

The most current nuclear magnetic resonance (NMR) instruments use a technique called pulsed Fourier transform NMR (FT NMR). In FT NMR, electromagnetic radiation excites all the ^{1}H or ^{13}C nuclei in a compound simultaneously, and this creates an oscillating sine curve called a free-induction delay (FID). The FID is then converted to a set of a frequencies via Fourier transform. Most studied organic compounds have many frequencies that contribute to their FIDs and require computer programming to be transformed to a spectrum.

Most modern NMR spectrometers operate at very high magnetic frequencies that can only be achieved using superconducting electromagnets. The materials required to form these magnets are only superconducting at very low temperatures, so they are stored in double-lined Dewar vessels and cooled by liquid helium and liquid nitrogen. There are many benefits to using higher-frequencied instruments. First, they have a greater sensitivity because of a greater difference in spin-state populations which results in stronger NMR signals. Additionally, there are larger energy differences between nuclei under higher frequency which results in greater signal separation.

Infrared Spectroscopy

Infrared (IR) spectroscopy provides an effective method for identifying the presence of common functional groups in a compound. When infrared energy is passed through an organic compound, absorption bands are observed that are correlated to the types of chemical bonds present in the structure of the compound, providing key information for identifying the functional groups present in the compound.

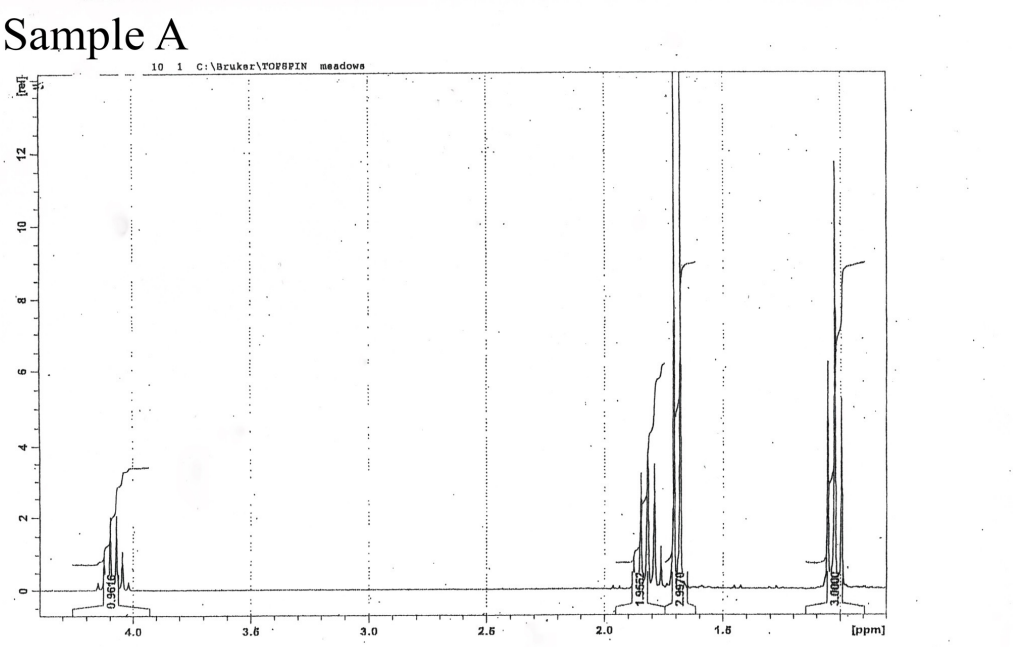
Most modern IR spectrometers use wavenumbers as the standard frequency scale for infrared radiation, and microcomputers incorporated into the instruments can quickly change between the two formats.

Gas Chromatograph-Mass Spectrometry

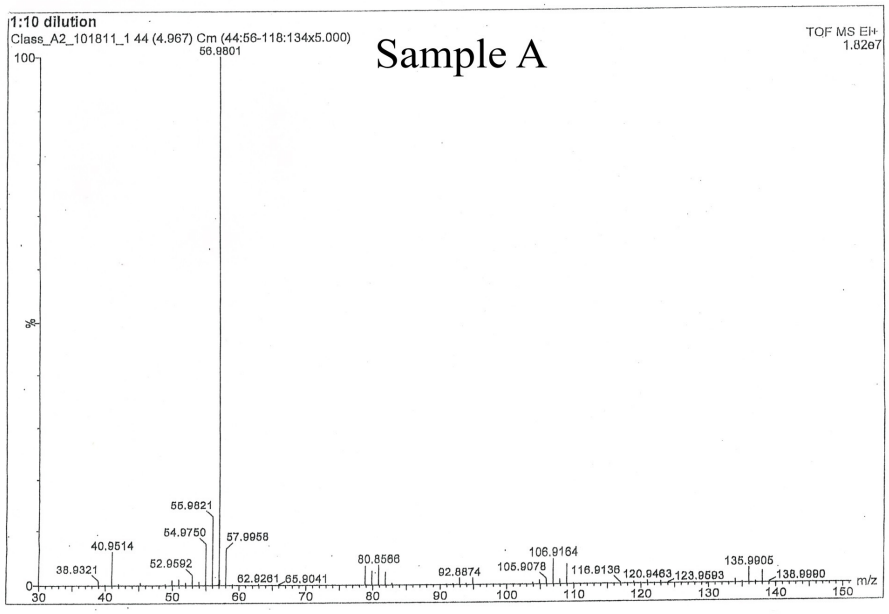
Many laboratories have instruments that combine and serve as a hybrid of a gas chromatograph and a mass spectrometer and are known as GCMS instruments. Small samples of a waste stream from a gas chromatograph and are directed into a mass spectrometer. The sample is then ionized by electron impact and the ions are passed into the mass analyzer, which in most cases is a quadrupole mass filter. As the ions passed through the central space, the combined direct and alternating currents cause them to oscillate, and the ions pass through the filter and strike the detector, resulting in a mass spectrum.

Data

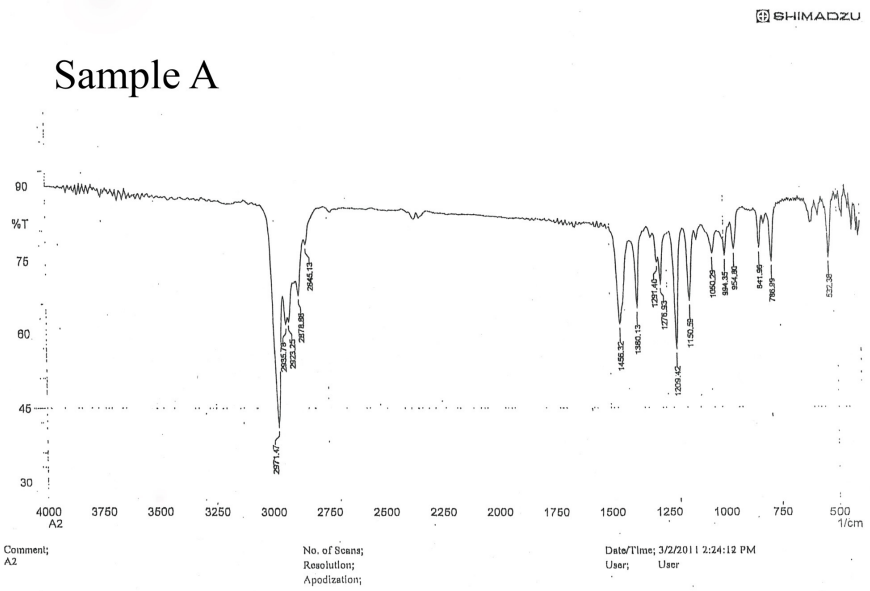
NMR Sample A



GCMS Sample A



FTIR Sample A



Analysis

Nuclear Magnetic Resonance Spectroscopy

The four unique signals found in the NMR spectrum indicate that there are four carbons in the organic compound. The integration of the peaks on the spectrum indicates that there are nine hydrogens in the compound. At a chemical shift value of approximately 4.1, there is a sextet, indicating that that carbon is part of an ethyl group. At δ1.9, there is a pentet, which is derived from the CH_2 carbon of the ethyl group. At the δ1.7, there is a doublet, which indicates that this area only has one hydrogen.

Infrared Spectroscopy

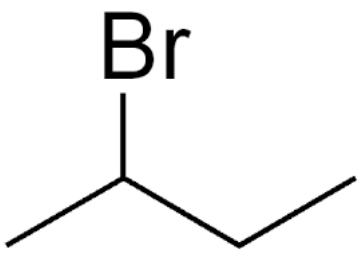
Based on the IR data, the largest peak is around 2971, which falls within the alkane range. In the given alkane range of 2850-3100, there are four dips that indicate there are four carbons present in the compound. The peak around 532 can be used to predict that there is a bromine atom present in the sample compound.

Mass Spectroscopy

The GCMS sample data shows that bromine is, indeed, present in the compound due to the characteristic 1:1 ratio of M^+/M+2. The isotopes ^{79}Br and ^{81}Br are found in equal abundance in nature. In this sample, M^+ is approximately 136 m/z and the base peak is present at 57 m/z, corresponding to the presence of a butyl (four carbon) group in the compound

Conclusion

Based on the three types of spectral analysis used, the compound must be 2-bromobutane. The compound has a chemical formula of C_4H_9Br, and the chemical structure is pictured below.



References

Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry 2 Lab Manual [lab manual]. Auburn (Al): Auburn University.